Design and implement an embedded access-control prototype that grants entry only when either a whitelisted RFID tag is detected or a legitimate fingerprint is verified, provides real-time visual feedback on an LCD, indicates status via an LED, and pulses a relay (active-LOW) to unlock for a fixed interval. The system must log all events (time, method, result) for auditability.

Installation Manual

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. RFID RC522
5. R307/R305 fingerprint sensor
6. 16×2 LCD

- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect with Raspberry Pi's GPIO Pins.**

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- **Relay Module**

Device / Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO23	16

- **LCD 16X2 Connections**

Device / Pin	Raspberry Pi 4	Physical Pin
LCD GND	GND	6
LCD VCC	5V	4
LCD SDA	GPIO2 / SDA	3
LCD SCL	GPIO3 / SCL	5

- **R307/R305 Fingerprint**

VCC	5V Pin 2	VCC
GND	GND Pin 6	GND
TX	GPIO15 / RXD, Pin 10	TX
RX	GPIO14 / TXD, Pin 8	RX

- **RC522 RFID → Raspberry Pi 4**

RC522 Pin	***Raspberry Pi Physical Pin***	***GPIO***
3.3V	***Pin 1***	***3.3V***
RST	**PIN 22**	**GPIO25**
GND	Pin 6	GND
MISO	Pin 21	GPIO9
MOSI	Pin 19	GPIO10
SCK	Pin 23	GPIO11
SDA/SS	Pin 24	GPIO8

Software Installation

sudo apt update
sudo apt install -y python3-pip python3-serial python3-rpi.gpio i2c-tools pip3 install RPLCD adafruit-circuitpython-fingerprint mfrc522

sudo raspi-conFig
Interface Options → I2C

→ Enable sudo i2cdetect

-y 1

I2C LCD Address

The LCD shown in the supplied image uses an I2C backpack. The common address is **0x27**. If i2cdetect shows
0x3F, change the LCD address in the program from 0x27 to 0x3F.

CODE:-

```
import RPi.GPIO as
GPIO import time
import csv
import os
import
serial
from datetime import
datetime from
RPLCD.i2c import
CharLCD from mfrc522
import SimpleMFRC522
import
adafruit_fingerprint

RELAY = 23
LED = 21

GPIO.setmode(GPIO.BC
M) GPIO.setup(RELAY,
GPIO.OUT)
GPIO.setup(LED,
GPIO.OUT)

# Active LOW relay
GPIO.output(RELAY,
GPIO.HIGH)
GPIO.output(LED,
GPIO.LOW)
```

```
lcd = CharLCD(
    i2c_expander='PCF857
    4', address=0x27,
    port=1,
    cols=16,
    rows=2,
    charmap='
    A00',
    auto_linebreaks=True
)

reader = SimpleMFRC522()

# Replace these with your actual
authorized RFID UIDs WHITELIST = [
    123456789,
    987654321
]

uart = serial.Serial("/dev/serial0", baudrate=57600,
timeout=1) finger =
adafruit_fingerprint.Adafruit_Fingerprint(uart)
LOG_FILE =
"access_log.csv" def
log_event(method,
result):
    new_file = not os.path.exists(LOG_FILE)
    with open(LOG_FILE, "a",
        newline="") as file: writer =
        csv.writer(file)
        if new_file:
            writer.writerow(["Time", "Method", "Result"])
        writer.writerow([
            datetime.now().strftime("%Y-%m-%d
            %H:%M:%S"), method,
            result
        ])

def show_message(line1,
    line2=""): lcd.clear()
    lcd.write_string(line1[:
    16]) if line2:
        lcd.cursor_pos = (1, 0)
        lcd.write_string(line2[:16
        ])

        def
unlock(method
        ):
    show_message("ACCESS
    GRANTED", method)
    GPIO.output(LED,
```

```
    GPIO.HIGH)
    GPIO.output(RELAY,
    GPIO.LOW)
    log_event(method,
    "GRANTED")
    time.sleep(5)
    GPIO.output(RELAY,
    GPIO.HIGH)
    GPIO.output(LED,
    GPIO.LOW)

def deny(method):
    show_message("ACCESS DENIED",
    method) GPIO.output(LED,
    GPIO.LOW)
    log_event(method,
    "DENIED") time.sleep(2)

def check_fingerprint():
    show_message("Place
    Finger") start =
    time.time()

    while time.time() - start <
        5: result =
        finger.get_image()

        if result ==
            adafruit_fingerprint.OK:
            break
        if result == adafruit_fingerprint.NOFINGER:
```

```
            time.sleep(0.1) continue

        return False
    else:
        return False

    show_message("Processing...")

    if finger.image_2_tz(1) != adafruit_fingerprint.OK:
        return False

    if finger.finger_search() !=
        adafruit_fingerprint.OK: return False

    return True

try:
    show_message("ACCESS CONTROL", "SYSTEM READY")
    time.sleep(2)

    while True:
        show_message("Scan RFID", "or

        Fingerprint") card_id, text =

        reader.read_no_block()

        if card_id:
            print("RFID UID:", card_id)

            if card_id in WHITELIST:
                unlock("RFID")
            else:
                deny("RFID")

            time.sleep(1)
            continue

        if check_fingerprint():
            print("Fingerprint matched")
            unlock("FINGERPRINT")
        time.sleep(1) except
KeyboardInterrupt:
    print("\nProgram stopped")

finally:
```

```
    GPIO.output(RELAY,
    GPIO.HIGH) GPIO.output(LED,
    GPIO.LOW)
    GPIO.cleanup()
    lcd.clear()
```

- **RFID UID Test Program**

```
from mfrc522 import
SimpleMFRC522 import
RPi.GPIO as GPIO
reader =
SimpleMFRC522() try:
    print("Place RFID card...")

    while True:
        uid, text = reader.read()
        print("RFID UID:", uid)
        print("Data:", text) break

finally:
    GPIO.cleanup()
```

- **Fingerprint Enrollment Program**

```
import serial
import adafruit_fingerprint

uart = serial.Serial("/dev/serial0", baudrate=57600,
timeout=1) finger =
adafruit_fingerprint.Adafruit_Fingerprint(uart)
location = int(input("Enter fingerprint ID (1-127): "))
print("Place finger...")
while finger.get_image() != adafruit_fingerprint.OK:
    pass

if finger.image_2_tz(1) !=
    adafruit_fingerprint.OK: print("Image
```

```
    conversion failed")
    exit()

print("Remove  finger")
input("Press Enter after removing finger...")

print("Place same finger again...")
while finger.get_image() !=
    adafruit_fingerprint.OK: pass

if finger.image_2_tz(2) !=
    adafruit_fingerprint.OK: print("Image
    conversion failed")
    exit()

if finger.create_model() !=
    adafruit_fingerprint.OK: print("Fingerprint
    mismatch")
    exit()

if finger.store_model(location) == adafruit_fingerprint.OK:
    print("Fingerprint enrolled successfully")
else:
    print("Enrollment  failed")
```

- ## Running the Programs

```
python3 rfid_test.py
python3 enroll.py
python3 access_control.py
```